Code : 100401/111403

B.Tech 4th Semester Exam., 2022

(New Course)

BIOLOGY FOR ENGINEERING

Time : 3 hours *Full Marks* : 70

Instructions :

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

*(iv) Question No. **1** is compulsory.*

1. Choose the correct answer of any *seven* of the following : 2×7=14

(a) RNA interference helps in

(i) cell proliferation

(ii) micropropagation

(iii) cell defence

(iv) cell differentiation

(b) Which one is found only in RNA?

(i) Cytosine

(ii) Adenine

(iii) Uracil

(iv) Guanine

(c) Restriction endonucleases are used in genetic engineering because

(i) they can degrade harmful proteins

(ii) they can join DNA fragments

(iii) they can cut DNA at specific base sequence

(iv) they can cut DNA at variable sites

(d) DNA is found in

(i) chromosomes

(ii) plastids

(iii) mitochondria

(iv) All of the above

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(e) Genetic engineering is

(i) plastic surgery

(ii) addition and removal of genes

(iii) study of extranuclear genes

(iv) All of the above

(f) The two strands of DNA are held together by the bonds of

(i) nitrogen

(ii) hydrogen

(iii) oxygen

(iv) carbon

(g) Transgenic plants are developed by

(i) introducing foreign gene

(ii) introducing gene mutations

(iii) deleting certain chromosome's parts

(iv) stopping spindle formation

(h) Which of the following is related to genetic engineering?

(i) Mitochondrion

(ii) Golgi apparatus

(iii) Lysosome

(iv) Plasmid

(i) The name of coronavirus is

(i) COVID-19

(ii) Crown

(iii) Coronaviridae

(iv) Halo

(j) The metal present in the chlorophyll is

(i) calcium

(ii) aluminium

(iii) zinc

(iv) magnesium

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2. Answer the following questions : 7+7=14

(a) Differentiate between the basic working mechanism of bird flying and aircraft flying.

(b) Write the basic differences between science and engineering.

3. Answer the following questions : 7+7=14

(a) Write Mendel's law of independent assortment.

(b) Discuss about the concept of epistasis.

4. What is the basic concept of recessive and dominance in law of inheritance? Discuss about single-gene disorders in humans. 14

5. Answer the following questions : 7+7=14

(a) Differentiate between amino acid and nucleic acid.

(b) Discuss about the basic classification of enzymes.

6. Write the differences between transcription and translation for protein synthesis in cell. 14

7. Answer the following questions : 7+7=14

(a) Differentiate between exothermic and endothermic reaction.

(b) Draw a flowchart of photosynthesis cycle in reference to synthesis of glucose in plant.

8. Write the methods about the identification and classification of microorganisms. 7+7=14

9. Write short notes on any *four* of the following : ˆ3½×4=14

(a) Autotrophs and Heterotrophs

(b) Genetic material

(c) Gene interaction

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(d) Evolution and Speciation

(e) Media composition

(f) Growth curve

★ ★ ★

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